Task 4.3: UNIQUE and PRIMARY KEY

1. Difference Between UNIQUE and PRIMARY KEY

PRIMARY KEY identifies each row in a table. It does not allow duplicate or NULL values. A table can have only one PRIMARY KEY.

UNIQUE does not allow duplicate values. In PostgreSQL, it can allow NULL values. A table can have many UNIQUE constraints.

Example:

```
CREATE TABLE users (     user_id
INTEGER PRIMARY KEY,
username TEXT UNIQUE,     email
TEXT UNIQUE
);
```

Here, every user has a different ID, username, and email.

2. Single-Column and Composite PRIMARY KEY

A single-column PRIMARY KEY uses one column to identify a row. It is simple and easy to use.

Example:

```
CREATE TABLE departments (
dept_id INTEGER PRIMARY KEY,
dept_name TEXT
);
```

A composite PRIMARY KEY uses two or more columns together. We use it when one column is not enough.

Example:

```
CREATE TABLE student_courses (
```

```
    student_id INTEGER,
course_id INTEGER,
    PRIMARY KEY (student_id, course_id)
);
```

This key does not allow the same student to take the same course twice in the table.

3. Why Can a Table Have Only One PRIMARY KEY but Many UNIQUE Constraints?

A table can have only one PRIMARY KEY because it is the main key for each row. It can use one column or several columns.

A table can have many UNIQUE constraints because different columns can have different rules.

For example, a users table can have a PRIMARY KEY for user_id, a UNIQUE constraint for username, and another UNIQUE constraint for email.

In conclusion, PRIMARY KEY identifies each row. UNIQUE prevents duplicate values.

Task 5.3: ON DELETE Options

1. ON DELETE RESTRICT

RESTRICT does not allow us to delete a category if products use this category.
For example, the Electronics category has two products: Laptop and Keyboard. If we try to delete this category, PostgreSQL shows an error.
The category cannot be deleted because products still use it.

2. ON DELETE CASCADE

CASCADE automatically deletes related rows.
For example, order 101 has two items. When we delete order 101, these two items are also deleted from the order_items table.
Order 102 and its items stay in the table.

3. Conclusion

RESTRICT protects data from accidental deletion. CASCADE deletes related data automatically.
Both options help us manage relationships between tables.